ia_study1_mainAnalyses

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```r
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
library(emmeans)
library(psych)
library(patchwork)
source("https://raw.githubusercontent.com/vyqlua/avi_functions/refs/heads/main/ipsatize_fn.R")
```

```r
sensitivity_analyses <- c("!is.na(subID)",
  "exclude_age != 1", "exclude_sd != 1",
  "exclude_age != 1 & exclude_sd != 1")
sensitivity_analyses_i <- c("!is.na(subID)", "exclude_age != 1")
```

Load Data

```
data <- read_csv("ia_study1_anon_data_computed.csv")
data <- data[ , -1]

levelBreakdown <- read_csv("ia_TrialsBreakdown.csv")
```

Are There Cultural Differences in Ideal Affect between European Americans and Taiwanese?

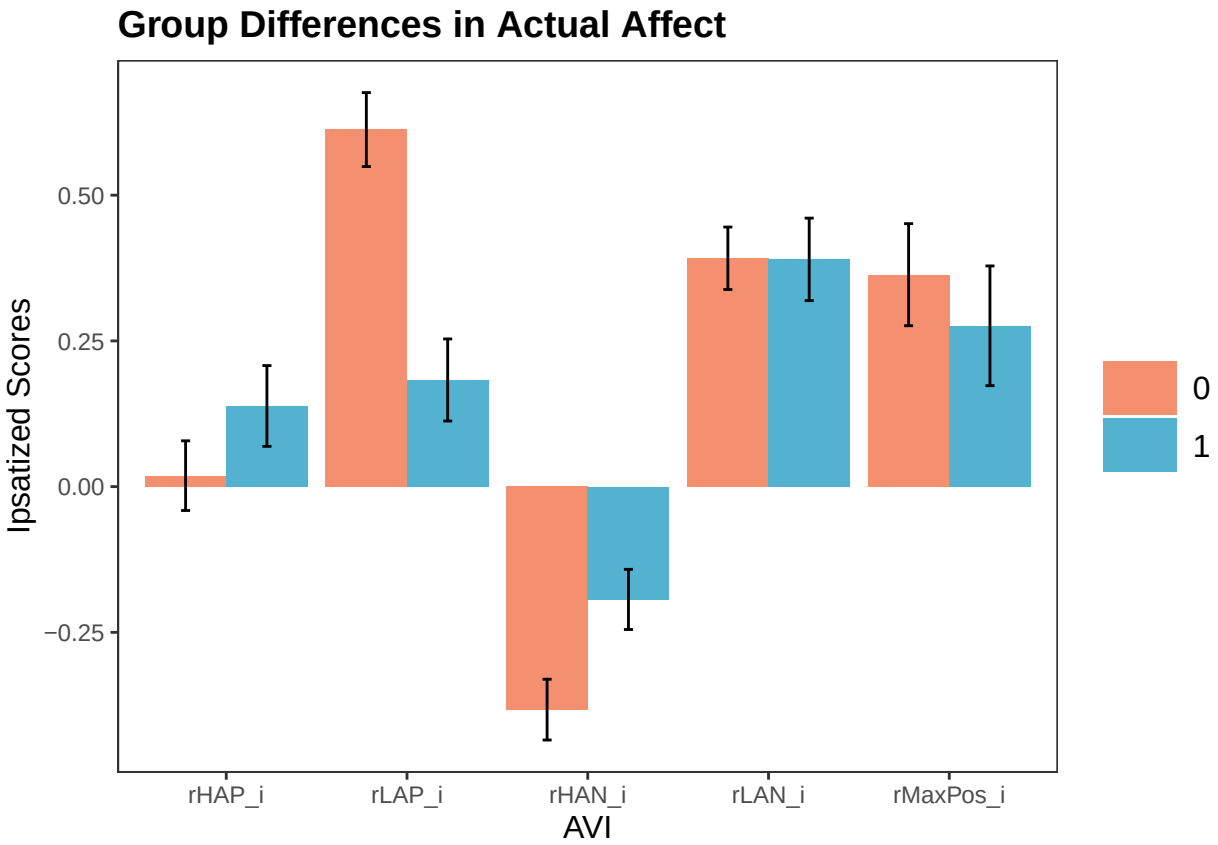
Visualization

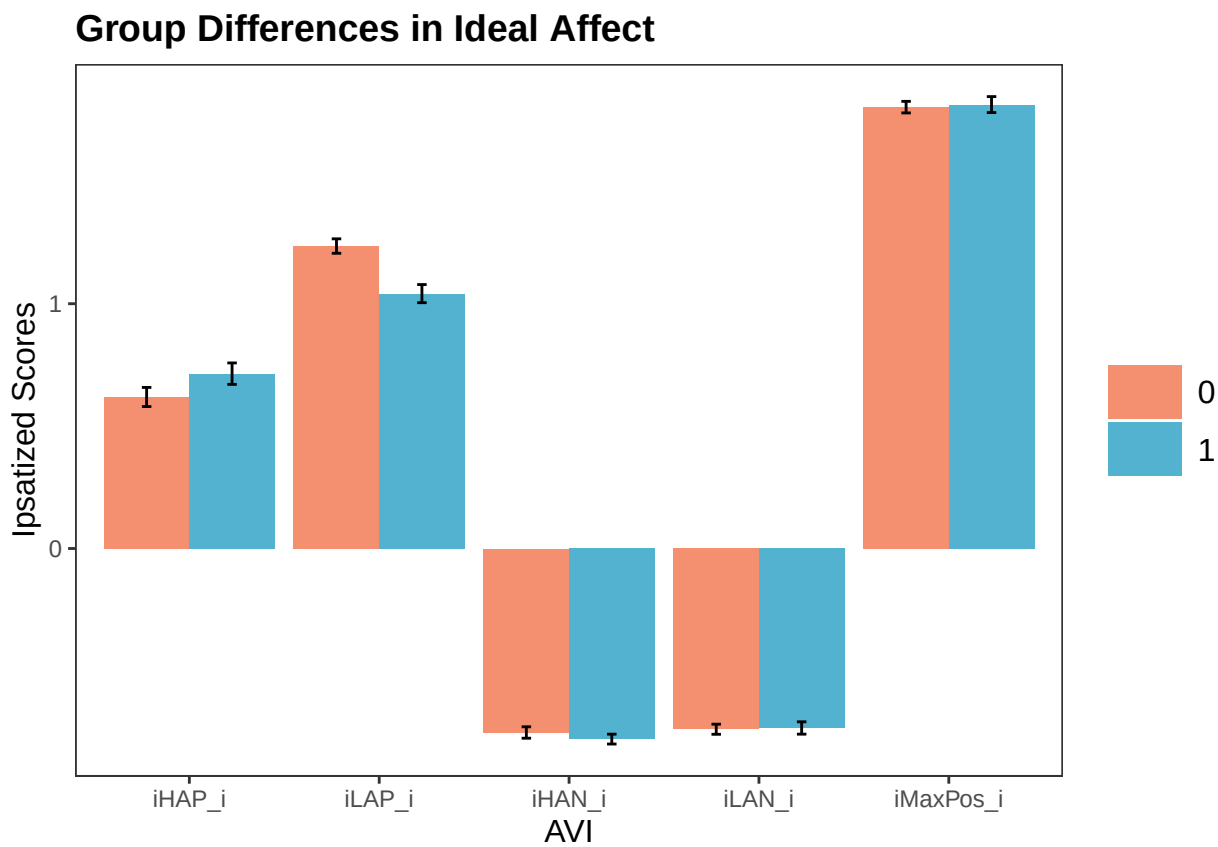
```r
plot_avi(data, group_id = "cultureEA")
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use ‘all_of()‘ or ‘any_of()‘ instead.
##   # Was:
##   data %>% select(group_id)
##
##   # Now:
##   data %>% select(all_of(group_id))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
## ‘summarise()‘ has grouped output by ’cultureEA’. You can override using the
## ‘.groups‘ argument.
```

```
## # A tibble: 10 x 6
## # Groups:   cultureEA [2]
##    cultureEA AVI          mean    sd     se AVI_order
##    <fct>     <fct>       <dbl> <dbl>  <dbl>     <dbl>
##  1 0         rHAP_i     0.0188 0.647 0.0598         1
##  2 1         rHAP_i     0.138  0.703 0.0693         1
##  3 0         rLAP_i     0.613  0.686 0.0634         3
##  4 1         rLAP_i     0.183  0.715 0.0704         3
##  5 0         rHAN_i    -0.383  0.564 0.0522         4
##  6 1         rHAN_i    -0.194  0.524 0.0516         4
##  7 0         rLAN_i     0.392  0.580 0.0536         6
##  8 1         rLAN_i     0.390  0.718 0.0708         6
##  9 0         rMaxPos_i  0.364  0.947 0.0876         7
## 10 1         rMaxPos_i  0.276  1.04  0.103          7
## # A tibble: 10 x 6
## # Groups:   cultureEA [2]
##    cultureEA AVI         mean    sd     se AVI_order
##    <fct>     <fct>      <dbl> <dbl>  <dbl>     <dbl>
##  1 0         iHAP_i     0.619 0.423 0.0391         1
##  2 1         iHAP_i     0.714 0.444 0.0438         1
##  3 0         iLAP_i     1.24  0.319 0.0295         3
##  4 1         iLAP_i     1.04  0.376 0.0370         3
##  5 0         iHAN_i    -0.752 0.252 0.0233         4
##  6 1         iHAN_i    -0.779 0.205 0.0202         4
##  7 0         iLAN_i    -0.738 0.223 0.0207         6
##  8 1         iLAN_i    -0.733 0.256 0.0252         6
##  9 0         iMaxPos_i  1.80  0.256 0.0237         7
## 10 1         iMaxPos_i  1.81  0.331 0.0326         7
```


Group Differences in Actual Affect
Ipsatized Scores
0.50
0.25
0.00
−0.25
rHAP_i
rLAP_i
rHAN_i
rLAN_i
rMaxPos_i
AVI
0
1



Analyses

```
# convert AVI data into long form
longavidata <- data %>%
  dplyr::select(
    subID, cultureEA, starts_with("exclude"),
    iHAP, iHAP_i, iLAP, iLAP_i,
    rHAP, rHAP_i, rLAP, rLAP_i) %>%
  pivot_longer(
    cols = `iHAP`:`iLAP_i`,
    names_to = "IdealAffect",
    values_to = "value")
```

```
# test if cultures differ on ideal affect (ipsatized scores)
idealAffectresult_i <-
  aov(value ~ IdealAffect*cultureEA + rHAP + rLAP + Error(subID/IdealAffect),
    data = longavidata %>%
      dplyr::filter(IdealAffect == "iHAP_i" |
                    IdealAffect == "iLAP_i"))
summary(idealAffectresult_i)
```

```
##
## Error: subID
```

```
##            Df Sum Sq Mean Sq F value   Pr(>F)
## cultureEA   1  0.266  0.2662   3.237   0.0734 .
## rHAP        1  2.271  2.2708  27.611 3.56e-07 ***
## rLAP        1  0.088  0.0881   1.071   0.3018
## Residuals 215 17.682  0.0822
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:IdealAffect
##                        Df Sum Sq Mean Sq F value Pr(>F)
## IdealAffect             1  25.41  25.413  117.96 <2e-16 ***
## IdealAffect:cultureEA   1   2.29   2.289   10.62 0.0013 **
## Residuals             217  46.75   0.215
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# pairwise tests (ipsatized scores)
idealHAPresult_i <-
  lm(value ~ cultureEA + rHAP,
     data = longavidata %>% dplyr::filter(IdealAffect == "iHAP_i"))
summary(idealHAPresult_i)
```

```
##
## Call:
## lm(formula = value ~ cultureEA + rHAP, data = longavidata %>%
##     dplyr::filter(IdealAffect == "iHAP_i"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.03012 -0.24617  0.02268  0.24275  1.02324
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.18022    0.09294  -1.939   0.0538 .
## cultureEA    0.06589    0.04989   1.321   0.1880
## rHAP         0.32805    0.03551   9.237   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3675 on 216 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.2918, Adjusted R-squared:  0.2853
## F-statistic:  44.5 on 2 and 216 DF,  p-value: < 2.2e-16
```

```r
idealLAPresult_i <-
  lm(value ~ cultureEA + rLAP,
     data = longavidata %>% dplyr::filter(IdealAffect == "iLAP_i"))
summary(idealLAPresult_i)
```

```
##
## Call:
## lm(formula = value ~ cultureEA + rLAP, data = longavidata %>%
##     dplyr::filter(IdealAffect == "iLAP_i"))
```

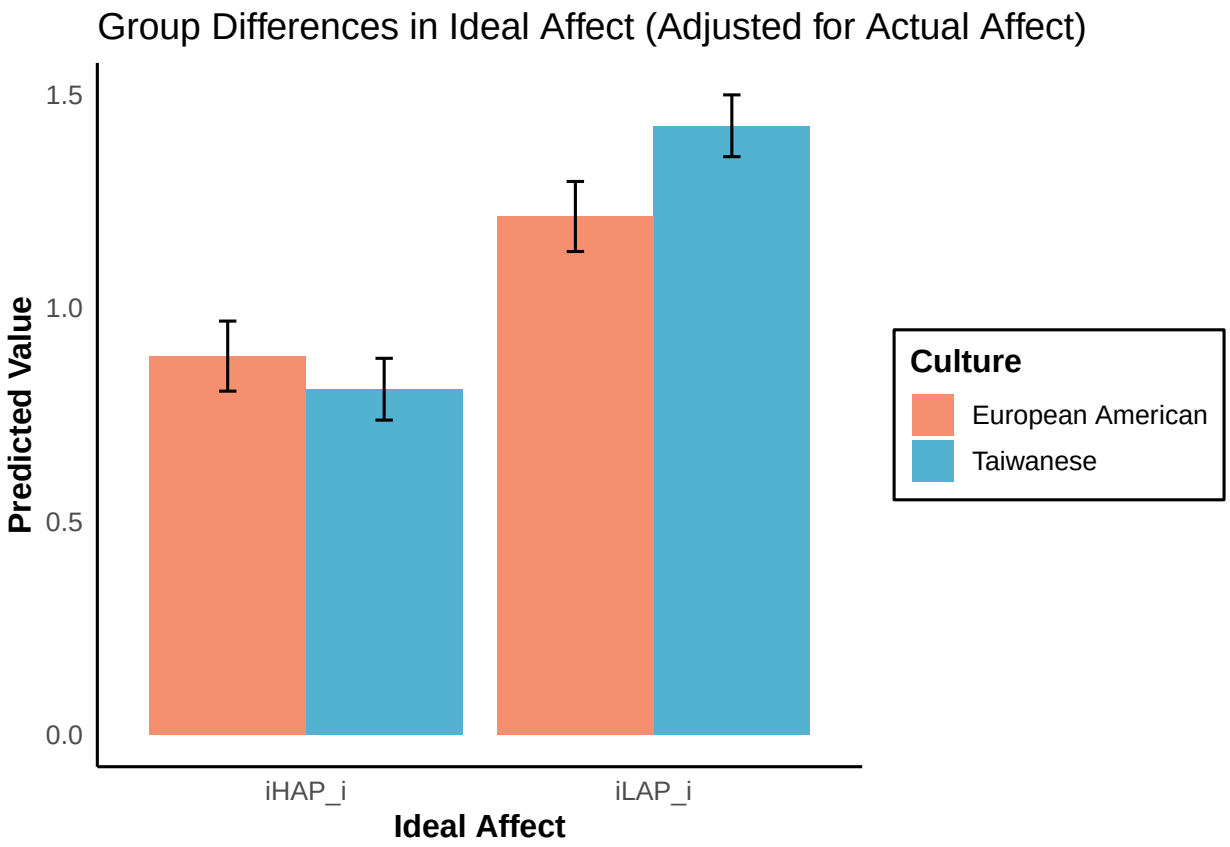
```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.14344 -0.23246  0.04333  0.22421  0.74581
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.09539    0.09819  11.156  < 2e-16 ***
## cultureEA   -0.17402    0.04873  -3.571 0.000439 ***
## rLAP         0.04641    0.03073   1.510 0.132395
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3457 on 216 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.0828, Adjusted R-squared:  0.07431
## F-statistic:  9.75 on 2 and 216 DF,  p-value: 8.833e-05
```

```r
emm <- emmeans(
  idealAffectresult_i,
  ~ IdealAffect * cultureEA
)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased
```

```r
plot_df <- as.data.frame(emm) %>%
  dplyr::mutate(
    Culture = if_else(cultureEA == 1, "European American", "Taiwanese") %>%
      as.factor()
  )

ggplot(plot_df %>%
         filter(grepl("_i", IdealAffect)),  # keep only the relevant IdealAffect levels
       aes(x = IdealAffect, y = emmean, fill = Culture)) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = emmean - SE, ymax = emmean + SE),
                position = position_dodge(0.9), width = 0.1) +
  scale_fill_manual(values = c(
    "European American"="#F49070",
    "Taiwanese"="#52B2CF")) +
  xlab("Ideal Affect") +
  ylab("Predicted Value") +
  ggtitle("Group Differences in Ideal Affect (Adjusted for Actual Affect)")
```


Group Differences in Ideal Affect (Adjusted for Actual Affect)
Predicted Value
1.5
1.0
0.5
0.0
iHAP_i
iLAP_i
Ideal Affect
Culture
European American
Taiwanese

Does Induced Mood Influence Giving Towards Excited, Calm, and Neutral Responders Differently Across Cultures?

Admin before analyses

```r
# Define the analysis function

aov_analysis <-
  function(formula, data, filter_condition1, filter_condition2 = NULL)
    {
    # Initialize a list to store results from each iteration
    analysis_result <- list()
  if (is.null(filter_condition2)) {
    for (i in 1:length(filter_condition1)) {
      subset_data <- data %>%
        filter(eval(parse(text = filter_condition1[i])))
    # Run the analysis on the subset
    analysis_result[[filter_condition1[i]]] <-
      aov(formula, data = subset_data) %>% summary()
    }}
  else {
    for (j in 1:length(filter_condition2)) {
      for (i in 1:length(filter_condition1)) {
        subset_data <- data %>%
          filter(eval(parse(text = filter_condition1[i]))) %>%
          filter(eval(parse(text = filter_condition2[j])))
    # Run the analysis on the subset
    analysis_result[[
      paste(filter_condition1[i], "&", filter_condition2[j])]] <-
      aov(formula = formula, data = subset_data) %>% summary()
      }}
    }
  return(analysis_result)
}
```

```r
# convert AVI data into long form
longGivingdata_i <- data %>%
  dplyr::select(
    subID, cultureEA, order,
    starts_with("raw") & contains("_i"),
    starts_with("exclude"),
    age # covariates
    ) %>%
  pivot_longer(
    cols = `raw1_i`:`raw36_i`,
    names_to = "trial",
    values_to = "response_i") %>%
  dplyr::mutate(
    trial = as.numeric(substr(trial, start = 4, stop = str_length(trial)-2)))
```

```r
# add info on incidental affect (mood), face gender, face race, and face emotion for each trial.
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    mood = levelBreakdown$mood[trial],
    faceGender = levelBreakdown$faceGender[trial],
    faceRace = levelBreakdown$faceRace[trial],
    faceEmotion = levelBreakdown$faceEmotion[trial])

# make variables factors
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    culture = ifelse(cultureEA == 1, "EA", "TN") %>% as.factor(),
    mood = factor(mood, levels = c("HAP", "HAN","Neutral")),
    faceRace = factor(faceRace, levels = c("White", "Asian")),
    faceGender = factor(faceGender, levels = c("Male", "Female")),
    faceEmotion = factor(faceEmotion, levels = c("Excited", "Calm", "Neutral")),
    subID = factor(subID, unique(subID)))

longGivingdata_i = longGivingdata_i %>% dplyr::filter(!is.na(response_i))
```

Sample size

Main analyses (!is.na(subID))

```r
print("n_total:"); longGivingdata_i %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 184
```

```r
print("n_tw:"); longGivingdata_i %>%
  dplyr::filter(culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 107
```

```r
print("n_ea:"); longGivingdata_i %>%
  dplyr::filter(culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 77
```

Supple analyses (exclude_age != 1)

```r
print("n_total:"); longGivingdata_i %>%
  dplyr::filter(exclude_age != 1) %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 175
```

```r
print("n_tw:"); longGivingdata_i %>%
  dplyr::filter(exclude_age != 1, culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 106
```

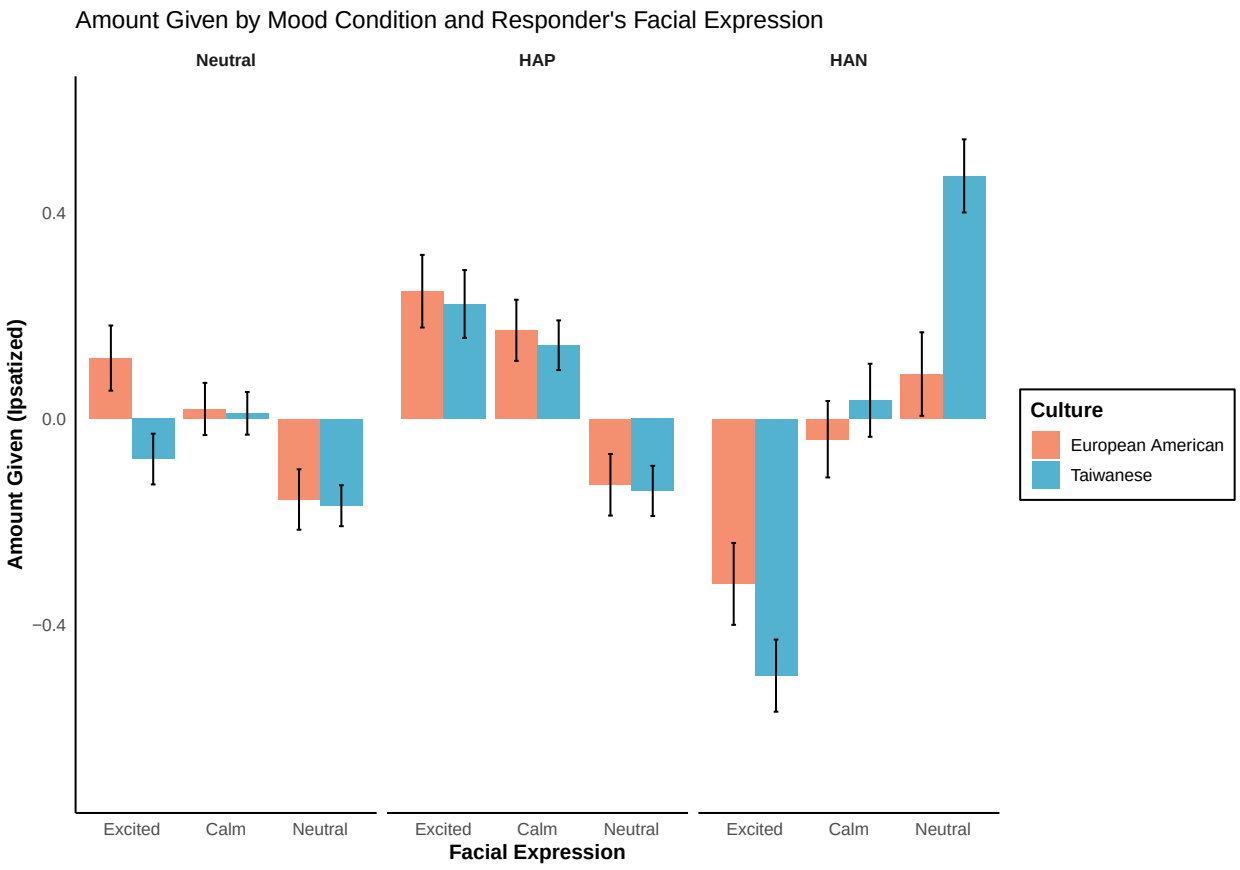
```r
print("n_ea:"); longGivingdata_i %>%
  dplyr::filter(exclude_age != 1, culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 69
```

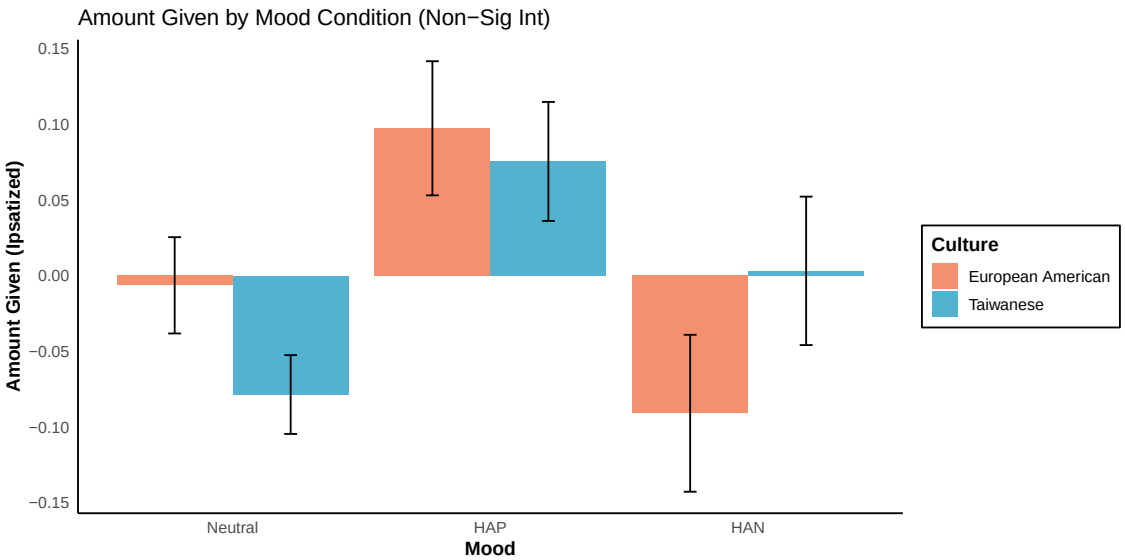
Visualizations

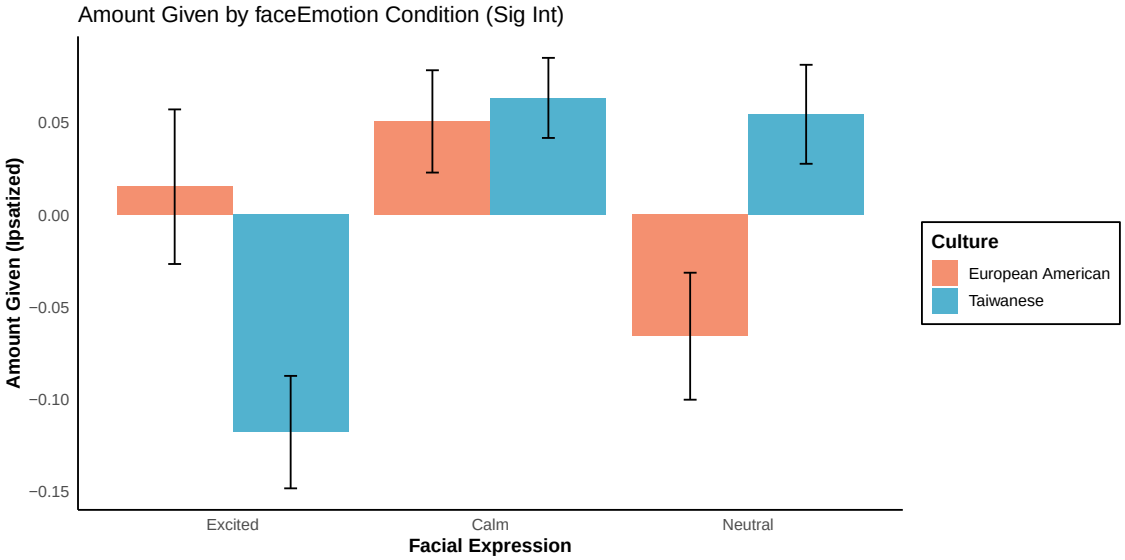
```
longGivingdata_i2 <- longGivingdata_i %>% dplyr::mutate(
  culture = if_else(culture == "EA", "European American", "Taiwanese")) %>%
  dplyr::rename(Culture = culture) %>%
  dplyr::mutate(
    mood = factor(mood,
                  labels = c("Neutral", "HAP", "HAN"),
                  levels = c("Neutral", "HAP", "HAN")))
ggplot(
  longGivingdata_i2 %>%
    group_by(subID, mood, faceEmotion, Culture) %>%
    summarise(response_i = mean(response_i, na.rm = T)) %>%
    group_by(faceEmotion, mood, Culture) %>%
    summarise(mean = mean(response_i, na.rm = T),
              SD = sd(response_i, na.rm = T),
              SE = sd(response_i, na.rm = T)/sqrt(n())),
  aes(faceEmotion, mean, fill = Culture)) +
  facet_grid(~mood)+
  scale_fill_manual(
    values = c("European American"="#F49070", "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1)+
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ylim(c(-0.7, 0.6)) +
  ggtitle("Amount Given by Mood Condition and Responder's Facial Expression")
```

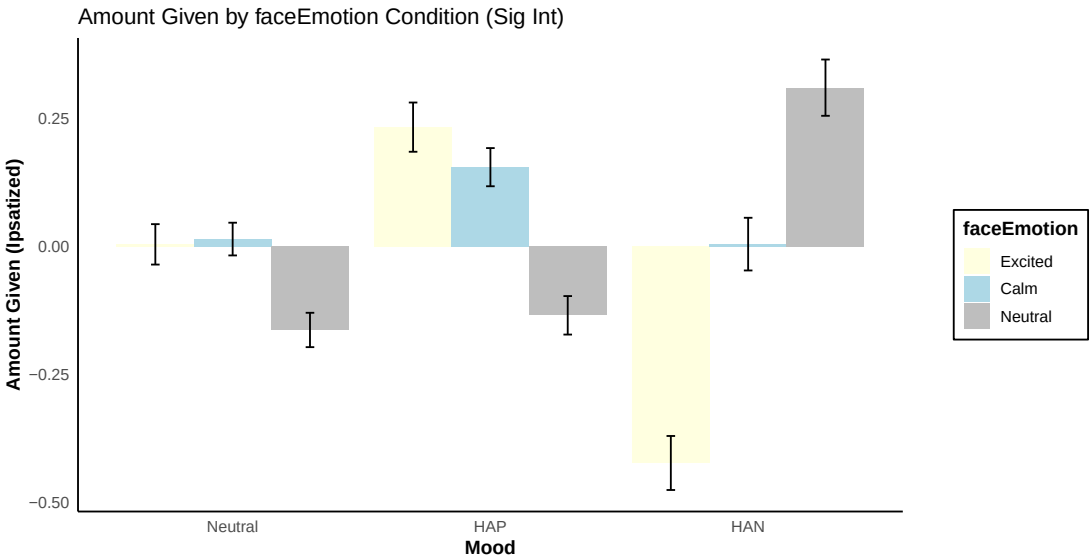

Amount Given by Mood Condition and Responder's Facial Expression
Neutral
HAP
HAN
0.4
0.0
−0.4
Amount Given (Ipsatized)
Excited
Calm
Neutral
Excited
Calm
Neutral
Excited
Calm
Neutral
Facial Expression
Culture
European American
Taiwanese

```r
p1 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, mood, Culture) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(mood, Culture) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by Mood Condition (Non-Sig Int)")

p2 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, faceEmotion, Culture) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(faceEmotion, Culture) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(faceEmotion, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition (Sig Int)")

p3 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, faceEmotion, mood) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(faceEmotion, mood) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = faceEmotion)) +
  scale_fill_manual(values =
                      c("Neutral"="grey",
                        "Excited"="lightyellow",
                        "Calm"="lightblue")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
```

```r
                 position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition (Sig Int)")

p1 + p2 + p3 +
  patchwork::plot_layout(
    ncol = 1
  )
```


Amount Given by Mood Condition (Non−Sig Int)
Amount Given (Ipsatized)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Neutral
HAP
HAN
Mood
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition (Sig Int)
Amount Given (Ipsatized)
0.05
0.00
−0.05
−0.10
−0.15
Excited
Calm
Neutral
Facial Expression
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition (Sig Int)
Amount Given (Ipsatized)
0.25
0.00
−0.25
−0.50
Neutral
HAP
HAN
Mood
faceEmotion
Excited
Calm
Neutral

Analyses

```r
aov_analysis(
  formula = response_i ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i,
  filter_condition1 = sensitivity_analyses_i)
```

```
## $`!is.na(subID)`
##
## Error: subID
##            Df    Sum Sq   Mean Sq F value Pr(>F)
## culture     1 7.000e-29 6.977e-29   0.318  0.573
## Residuals 182 3.987e-26 2.191e-28
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   23.9  11.926   4.227 0.0153 *
## mood:culture   2    7.8   3.903   1.384 0.2520
## Residuals    364 1027.0   2.821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## faceEmotion           2   16.0   7.993   5.303 0.00537 **
## culture:faceEmotion   2   17.4   8.680   5.759 0.00345 **
## Residuals           364  548.6   1.507
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value  Pr(>F)
## mood:faceEmotion           4  254.5   63.63  64.318 < 2e-16 ***
## mood:culture:faceEmotion   4   15.2    3.81   3.852 0.00418 **
## Residuals                728  720.2    0.99
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 4968   3810  0.7668
##
## $`exclude_age != 1`
##
## Error: subID
##            Df    Sum Sq   Mean Sq F value Pr(>F)
## culture     1 1.800e-28 1.784e-28   0.339  0.561
## Residuals 173 9.109e-26 5.266e-28
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   23.7  11.867   4.228 0.0153 *
```

```
## mood:culture   2    4.5   2.240   0.798 0.4511
## Residuals    346  971.2   2.807
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## faceEmotion           2   19.1   9.561   6.161 0.00235 **
## culture:faceEmotion   2   14.1   7.073   4.558 0.01112 *
## Residuals           346  537.0   1.552
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value  Pr(>F)
## mood:faceEmotion           4  242.5   60.61  61.089 < 2e-16 ***
## mood:culture:faceEmotion   4   15.9    3.98   4.014 0.00316 **
## Residuals                692  686.6    0.99
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 4725   3610  0.7641
```

Pairwise

```
pairwise <- aov(
  formula = response_i ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i)
```

Main effect mood

```r
# main effect mood
emm_3way_mood <- emmeans(pairwise, ~ mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts

## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_mood) # p values
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  HAP - HAN       0.1302 0.0512 364   2.542  0.0307
##  HAP - Neutral   0.1288 0.0512 364   2.515  0.0330
##  HAN - Neutral  -0.0014 0.0512 364  -0.027  0.9996
##
## Results are averaged over the levels of: culture, faceEmotion
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_mood)) # confidence intervals
```

Main effect faceEmotion

```r
# main effect faceEmotion
emm_3way_faceEmotion <- emmeans(pairwise, ~ faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts

## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotion) # p values
```

```
##  contrast          estimate     SE  df t.ratio p.value
##  Excited - Calm     -0.1082 0.0375 364  -2.890  0.0113
##  Excited - Neutral  -0.0456 0.0375 364  -1.218  0.4434
##  Calm - Neutral      0.0626 0.0375 364   1.672  0.2172
##
## Results are averaged over the levels of: mood, culture
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_faceEmotion)) # confidence intervals
```

Culture:faceEmotion

```r
emm_3way_faceEmotionCulture <- emmeans(pairwise, ~ culture | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionCulture) # p values
```

```
## faceEmotion = Excited:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN    0.1330 0.0432 364   3.075  0.0023
##
## faceEmotion = Calm:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN   -0.0127 0.0432 364  -0.295  0.7685
##
## faceEmotion = Neutral:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN   -0.1202 0.0432 364  -2.781  0.0057
##
## Results are averaged over the levels of: mood
```

```r
#confint(pairs(emm_3way_faceEmotionCulture)) # confidence intervals
```

Mood:faceEmotion

```r
emm_3way_faceEmotionMood <- emmeans(pairwise, ~ faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionMood) # p values
```

```
## mood = HAP:
##  contrast          estimate    SE   df t.ratio p.value
##  Excited - Calm     0.07800 0.057 1046   1.370  0.3574
##  Excited - Neutral  0.36934 0.057 1046   6.485  <.0001
##  Calm - Neutral     0.29135 0.057 1046   5.116  <.0001
##
## mood = HAN:
##  contrast          estimate    SE   df t.ratio p.value
##  Excited - Calm    -0.40767 0.057 1046  -7.158  <.0001
##  Excited - Neutral -0.68870 0.057 1046 -12.093  <.0001
##  Calm - Neutral    -0.28104 0.057 1046  -4.935  <.0001
##
## mood = Neutral:
##  contrast          estimate    SE   df t.ratio p.value
```

```
##  Excited - Calm     0.00495 0.057 1046   0.087  0.9958
##  Excited - Neutral  0.18254 0.057 1046   3.205  0.0040
##  Calm - Neutral     0.17759 0.057 1046   3.118  0.0053
##
## Results are averaged over the levels of: culture
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#confint(pairs(emm_3way_faceEmotionMood)) # confidence intervals
```

Mood:culture:faceEmotion

```r
emm_3way_bymood <- emmeans(pairwise, ~ culture * faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_bymood) # p values
```

```
## mood = HAP:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.02486 0.0885 1188   0.281  0.9998
##  EA Excited - EA Calm     0.07592 0.0869 1046   0.874  0.9527
##  EA Excited - TN Calm     0.10493 0.0885 1188   1.186  0.8436
##  EA Excited - EA Neutral  0.37585 0.0869 1046   4.327  0.0002
##  EA Excited - TN Neutral  0.38770 0.0885 1188   4.383  0.0002
##  TN Excited - EA Calm     0.05106 0.0885 1188   0.577  0.9925
##  TN Excited - TN Calm     0.08007 0.0737 1046   1.087  0.8868
##  TN Excited - EA Neutral  0.35099 0.0885 1188   3.968  0.0011
##  TN Excited - TN Neutral  0.36284 0.0737 1046   4.924  <.0001
##  EA Calm - TN Calm        0.02901 0.0885 1188   0.328  0.9995
##  EA Calm - EA Neutral     0.29993 0.0869 1046   3.453  0.0076
##  EA Calm - TN Neutral     0.31178 0.0885 1188   3.525  0.0059
##  TN Calm - EA Neutral     0.27092 0.0885 1188   3.063  0.0272
##  TN Calm - TN Neutral     0.28277 0.0737 1046   3.838  0.0018
##  EA Neutral - TN Neutral  0.01185 0.0885 1188   0.134  1.0000
##
## mood = HAN:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.17807 0.0885 1188   2.013  0.3354
##  EA Excited - EA Calm    -0.28085 0.0869 1046  -3.233  0.0159
##  EA Excited - TN Calm    -0.35642 0.0885 1188  -4.029  0.0008
##  EA Excited - EA Neutral -0.40735 0.0869 1046  -4.690  <.0001
##  EA Excited - TN Neutral -0.79199 0.0885 1188  -8.953  <.0001
##  TN Excited - EA Calm    -0.45892 0.0885 1188  -5.188  <.0001
##  TN Excited - TN Calm    -0.53449 0.0737 1046  -7.254  <.0001
##  TN Excited - EA Neutral -0.58542 0.0885 1188  -6.618  <.0001
##  TN Excited - TN Neutral -0.97006 0.0737 1046 -13.165  <.0001
##  EA Calm - TN Calm       -0.07557 0.0885 1188  -0.854  0.9570
##  EA Calm - EA Neutral    -0.12650 0.0869 1046  -1.456  0.6922
##  EA Calm - TN Neutral    -0.51114 0.0885 1188  -5.778  <.0001
##  TN Calm - EA Neutral    -0.05093 0.0885 1188  -0.576  0.9926
##  TN Calm - TN Neutral    -0.43557 0.0737 1046  -5.911  <.0001
##  EA Neutral - TN Neutral -0.38464 0.0885 1188  -4.348  0.0002
##
## mood = Neutral:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.19603 0.0885 1188   2.216  0.2310
##  EA Excited - EA Calm     0.09879 0.0869 1046   1.137  0.8658
##  EA Excited - TN Calm     0.10714 0.0885 1188   1.211  0.8315
##  EA Excited - EA Neutral  0.27454 0.0869 1046   3.161  0.0201
##  EA Excited - TN Neutral  0.28658 0.0885 1188   3.240  0.0155
##  TN Excited - EA Calm    -0.09724 0.0885 1188  -1.099  0.8818
##  TN Excited - TN Calm    -0.08889 0.0737 1046  -1.206  0.8339
```

```
##  TN Excited - EA Neutral  0.07851 0.0885 1188   0.888  0.9495
##  TN Excited - TN Neutral  0.09055 0.0737 1046   1.229  0.8227
##  EA Calm - TN Calm        0.00835 0.0885 1188   0.094  1.0000
##  EA Calm - EA Neutral     0.17575 0.0869 1046   2.023  0.3296
##  EA Calm - TN Neutral     0.18779 0.0885 1188   2.123  0.2761
##  TN Calm - EA Neutral     0.16740 0.0885 1188   1.892  0.4073
##  TN Calm - TN Neutral     0.17944 0.0737 1046   2.435  0.1451
##  EA Neutral - TN Neutral  0.01204 0.0885 1188   0.136  1.0000
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```r
#confint(pairs(emm_3way_bymood)) # confidence intervals
```

```r
emm_3way_byfaceEmotion <- emmeans(pairwise, ~ culture * mood | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_byfaceEmotion) # p values
```

```
## faceEmotion = Excited:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.02486 0.0885 1188   0.281  0.9998
##  EA HAP - EA HAN          0.56821 0.1019  846   5.575  <.0001
##  EA HAP - TN HAN          0.74628 0.0885 1188   8.436  <.0001
##  EA HAP - EA Neutral      0.12987 0.1019  846   1.274  0.7991
##  EA HAP - TN Neutral      0.32590 0.0885 1188   3.684  0.0033
##  TN HAP - EA HAN          0.54335 0.0885 1188   6.142  <.0001
##  TN HAP - TN HAN          0.72142 0.0865  846   8.343  <.0001
##  TN HAP - EA Neutral      0.10501 0.0885 1188   1.187  0.8431
##  TN HAP - TN Neutral      0.30103 0.0865  846   3.482  0.0069
##  EA HAN - TN HAN          0.17807 0.0885 1188   2.013  0.3354
##  EA HAN - EA Neutral     -0.43834 0.1019  846  -4.300  0.0003
##  EA HAN - TN Neutral     -0.24231 0.0885 1188  -2.739  0.0685
##  TN HAN - EA Neutral     -0.61641 0.0885 1188  -6.968  <.0001
##  TN HAN - TN Neutral     -0.42038 0.0865  846  -4.862  <.0001
##  EA Neutral - TN Neutral  0.19603 0.0885 1188   2.216  0.2310
##
## faceEmotion = Calm:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.02901 0.0885 1188   0.328  0.9995
##  EA HAP - EA HAN          0.21144 0.1019  846   2.074  0.3017
##  EA HAP - TN HAN          0.13587 0.0885 1188   1.536  0.6410
##  EA HAP - EA Neutral      0.15274 0.1019  846   1.498  0.6654
##  EA HAP - TN Neutral      0.16109 0.0885 1188   1.821  0.4525
##  TN HAP - EA HAN          0.18243 0.0885 1188   2.062  0.3080
##  TN HAP - TN HAN          0.10686 0.0865  846   1.236  0.8191
##  TN HAP - EA Neutral      0.12373 0.0885 1188   1.399  0.7279
##  TN HAP - TN Neutral      0.13208 0.0865  846   1.528  0.6465
##  EA HAN - TN HAN         -0.07557 0.0885 1188  -0.854  0.9570
##  EA HAN - EA Neutral     -0.05870 0.1019  846  -0.576  0.9926
##  EA HAN - TN Neutral     -0.05035 0.0885 1188  -0.569  0.9930
##  TN HAN - EA Neutral      0.01687 0.0885 1188   0.191  1.0000
```

```
##  TN HAN - TN Neutral      0.02522 0.0865  846   0.292  0.9997
##  EA Neutral - TN Neutral  0.00835 0.0885 1188   0.094  1.0000
##
## faceEmotion = Neutral:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.01185 0.0885 1188   0.134  1.0000
##  EA HAP - EA HAN         -0.21499 0.1019  846  -2.109  0.2833
##  EA HAP - TN HAN         -0.59963 0.0885 1188  -6.779  <.0001
##  EA HAP - EA Neutral      0.02856 0.1019  846   0.280  0.9998
##  EA HAP - TN Neutral      0.04060 0.0885 1188   0.459  0.9975
##  TN HAP - EA HAN         -0.22684 0.0885 1188  -2.564  0.1070
##  TN HAP - TN HAN         -0.61148 0.0865  846  -7.072  <.0001
##  TN HAP - EA Neutral      0.01671 0.0885 1188   0.189  1.0000
##  TN HAP - TN Neutral      0.02875 0.0865  846   0.332  0.9995
##  EA HAN - TN HAN         -0.38464 0.0885 1188  -4.348  0.0002
##  EA HAN - EA Neutral      0.24355 0.1019  846   2.389  0.1610
##  EA HAN - TN Neutral      0.25559 0.0885 1188   2.889  0.0453
##  TN HAN - EA Neutral      0.62819 0.0885 1188   7.101  <.0001
##  TN HAN - TN Neutral      0.64023 0.0865  846   7.404  <.0001
##  EA Neutral - TN Neutral  0.01204 0.0885 1188   0.136  1.0000
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
#confint(pairs(emm_3way_byfaceEmotion)) # confidence intervals
```